Anna Chrzanowska

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Summary

- Experienced neuroscientist with 10+ years of expertise in behavioral systems neuroscience, specializing in rodent behavioral assays, neural circuit genetic manipulations, and neural activity recordings (*in vivo* freely-moving and head-fixed electrophysiology, two-photon calcium imaging). Skilled in experimental setup design, troubleshooting, and microsurgeries. Successful in securing fellowships and competitive training programs. Active in mentoring, organizing symposia and conferences, science popularization, and the international community of young neuroscientists.

Skills

Technical: Rodent behavioral assays, neural circuit genetic manipulations, *in vivo* electrophysiology (freely-moving & head-fixed), two-photon calcium imaging, microsurgeries, brain histology, viral injections, optogenetics, chemogenetics, patch clamp recordings (*in vitro*).

Engineering & Software: Experimental setup design, optimization, troubleshooting, Bonsai reactive programming, Arduino, electronics, virtual reality environments for mice.

Professional: Mentoring, project supervision, scientific writing, manuscript editing, organizing symposia and conferences, science popularization.

Education

PhD in Biology, KU Leuven (2018–2023): Arenberg Doctoral School, Leuven, Belgium. Dissertation: *Unraveling the cell-type specific underpinnings that shape collicular-mediated visual behaviors*. Supervisor: Dr. Karl Farrow (NERF).

MSc in Neurobiology, Jagiellonian University (2015–2017): Cracow, Poland. Dissertation: *Circadian analysis of the orexin fibers distribution in the rat lateral geniculate nucleus of the thalamus*. Supervisor: Prof. Dr. Marian H. Lewandowski.

BSc in Neurobiology, Jagiellonian University (2012–2015): Cracow, Poland. Dissertation: *Role of GABA-ergic innervation in the mammalian biological clock*. Supervisor: Prof. Dr. Marian H. Lewandowski.

Research Experience

Postdoctoral Researcher — ICM, Paris Brain Institute (2024–present): Bassem Hassan Lab, Paris, France. Investigating links between brain wiring structure, physiology, and behavior. *In vivo* two-photon calcium imaging in behaving flies; neurodevelopmental fly genetics.

Visiting Scientist — University of Mainz (2025): Marion Silies Lab, Mainz, Germany. Data analysis of calcium imaging signals.

PhD Researcher — NERF, imec / KU Leuven / VIB (2018–2023): Karl Farrow Lab, Leuven, Belgium. Role of distinct neuronal cell types in visually-guided innate behaviors. *In vivo* freely-moving and head-fixed Neuropixels recordings; optogenetics, chemogenetics, viral injections.

Undergraduate Researcher — Jagiellonian University (2012–2017): Marian H. Lewandowski Lab, Cracow, Poland. Circadian changes of orexinergic innervation in the rat visual system; brain histology; *in vitro* patch clamp.

Research Intern — NERF (2017): Karl Farrow Lab, Leuven, Belgium. Behavioral protocol for visually-triggered orienting and defensive behaviors in virtual reality.

Research Intern — Leibniz-Institute for Molecular Pharmacology, FMP (2016–2017): Ponomarenko / Korotkova Lab, Berlin, Germany. Sleep and memory regulation by forebrain circuits; optogenetics; *in vivo* local field potential recordings.

Research Intern — Intl. Institute of Molecular and Cell Biology (2016): Kuźnicki Lab, Warsaw, Poland. *In situ* proximity ligation assay (PLA) on brain slices for Huntington's disease research.

Erasmus+ Trainee — Neuroscience Institute, CNR (2015): Matteoli Lab, Milan, Italy. Role of Eps8 in synapse plasticity; Western blot analysis.

Awards and Distinctions

- Carnot Training Grant for short collaborative visits (2025)
- Marie Skłodowska-Curie Actions Postdoctoral Fellowship (2025)
- French Foundation for Medical Research (FRM) Postdoctoral Fellowship (2024)
- Fyssen Foundation Postdoctoral Fellowship (2023)
- Transylvanian Experimental Neuroscience Summer School (TENSS), participant (2021)
- CAJAL NeuroKit Course, participant (2021)
- Research Foundation – Flanders (FWO) PhD Fellowship, ~20% success rate (2018–2022)
- Travel Grant — CAJAL Advanced Neuroscience Training Programme (2018)
- Travel Grant — 7th Neuron Technology Summer School (2017)
- Deans' Award for Outstanding Academic Achievement (2015/16, 2016/17)
- Laureate — Grasz o staż (Win the Internship) competition (2016)
- Best Theoretical Poster — 5th Aspects of Neuroscience (2015)
- Best Theoretical Poster — 4th Aspects of Neuroscience (2014)

Publications

- **Chrzanowska A**, Kostecki M, Krasilshchikova N, Perrino M, Petrucco L. *Rethinking naturalistic behavior: The Terzolas Manifesto*. The Open Science Framework (2025). doi:10.31219/osf.io/ch3an_v2
- Londoño-Ramírez H, Huang X, Cools J, **Chrzanowska A**, et al. *Multiplexed surface electrode arrays based on metal oxide thin-film electronics for high-resolution cortical mapping*. Advanced Science (2023). doi:10.1002/advs.202308507
- Sans-Dublanc A, **Chrzanowska A**, Reinhard K, et al. *Optogenetic fUSI for brain-wide mapping of neural activity mediating collicular-dependent behaviors*. Neuron (2021). doi:10.1016/j.neuron.2021.04.008 — equal contribution, co-lead author
- Canti L, **Chrzanowska A**, Doglio MG, Martina L, Van Den Bossche T. *Research culture: science from bench to society*. Biology Open (2021). doi:10.1242/bio.058919 — all authors contributed equally
- Chrobok L, Palus K, **Chrzanowska A**, Kępczyński M, Lewandowski M. *Multiple excitatory actions of orexins upon thalamo-cortical neurons in dorsal lateral geniculate nucleus*. Scientific Reports (2017). doi:10.1038/s41598-017-08202-8
- Chrobok L, Palus K, Jeczmien-Lazur JS, **Chrzanowska A**, Kępczyński M, Lewandowski M. *Disinhibition of the intergeniculate leaflet network in the WAG/Rij rat model of absence epilepsy*. Experimental Neurology (2017). doi:10.1016/j.expneurol.2016.12.014
- Nuttin B, **Chrzanowska A**, Bruynsteen A, Sans-Dublanc A, Farrow K. *The role of collicular cell types in behavioral reactions to different threat intensities in mice*. In preparation — equal contribution, co-lead author.

Presentations

- **Invited Speaker** — How to write a successful MSCA proposal. PORT for Health Neuroscience, Wrocław, Poland (2025)
- **Invited Speaker** — Cell vibes: how collicular cell-types contribute to guiding mice innate behaviors. Neurons in Action, Warsaw, Poland (2025)
- **Oral** — Towards embracing brain complexity: lessons from cell types of the superior colliculus. ENCODS, Paris, France (2022)
- **Oral** — A cell-type specific understanding of innate defensive behaviors: the collicular perspective. Neuromatch 3.0 (2020)
- **Poster** — Neural circuits of individuality in the context of learning. VIP & DIF, Paris, France (2024)
- **Poster** — Locomotion shapes visually-evoked neural dynamics through modulation of collicular populations. COSYNE, Lisbon, Portugal (2024)

Student Supervision

Master's Students: Helin Polat, Hala Rachel El Hajj (2024–2025); Chaimae Ouriaghli Taouil (2021–2022); Liuyin Yang, Nidaa Elsayed (2019–2020); Carolina Tigre Avelar (2017–2018)

Bachelor's Students: Sofia Rebiha Decesaro (2024); Diana Dayekh (2023–2024); Martyna Palys, Sade Adeyemi (2022–2023); Diethe Peeters, Tom Voet, Mathis Van den Bruel (2021–2022); Mathilda Corcoles (2020–2021)

Major Collaborations

- **Alan Urban, NERF** — established novel technology to study cell-type specific brain-wide circuits in mice (opto-fUSI).
- **Nanoengineering team, imec** — tested *in vivo* a novel device for large-scale brain recordings.

Scientific Community

- Teaching assistant, Transatlantic Behavioral Neuroscience Summer School (2021 – present)
- Local teaching assistant, CAJAL NeuroKit Course at NERF (2022)